

Part A: Target Selection

Website Name: National Informatics Centre

URL: [National Informatics Centre](https://www.nic.in)

Reason for choosing the target:

I selected this government website because it is a public-facing website that provides official information and is appropriate for academic learning about website security and accessibility.

Part B: WHOIS Lookup

Field	Value
Website Name	National Informatics Centre (NIC)
URL	https://www.nic.in
Domain Name	nic.in
Registrar	National Informatics Centre
Registration Date	23 December 2004
Expiry Date	23 December 2026
Name Servers	ns6.nic.in, ns8.nic.in, nicnet.nic.in, ns4.nic.in
Domain Status	ok

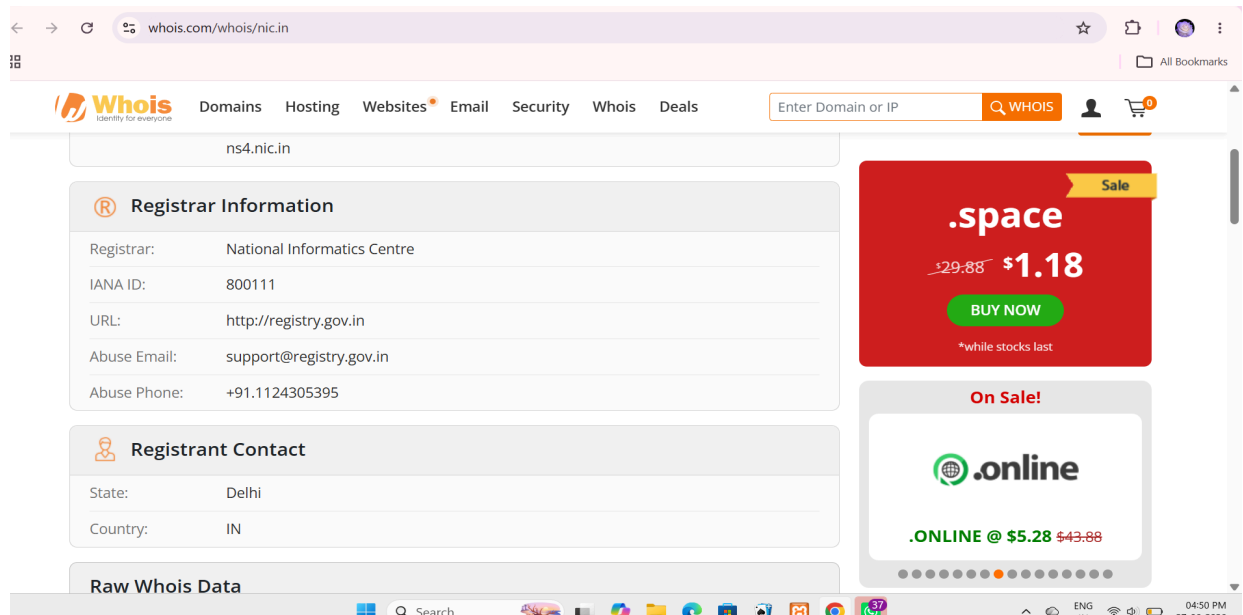
The screenshot shows a web browser displaying the WHOIS lookup results for the domain **nic.in** on the website whois.com/whois/nic.in. The page features a navigation bar with links for Domains, Hosting, Websites, Email, Security, Whois, and Deals. A search bar with the text "Enter Domain or IP" and a "WHOIS" button is present. The main content area displays the domain **nic.in** with a note "Updated 3 days ago". Below this, there is a "Domain Information" section with the following details:

Domain:	nic.in
Registered On:	2004-12-23
Expires On:	2026-12-23
Updated On:	2026-02-06
Status:	ok
Name Servers:	ns6.nic.in ns8.nic.in nicnet.nic.in ns4.nic.in

Below the domain information is a "Registrar Information" section. To the right of the domain information, there is a section titled "Interested in similar domains?" with a list of suggested domains, each with a "Buy Now" button:

- [nicphotography.com](#) Buy Now
- [nicassociates.com](#) Buy Now
- [nicttravel.com](#) Buy Now
- [thenicfamily.com](#) Buy Now
- [nicdesign.net](#) Buy Now
- [nicfamily.net](#) Buy Now

At the bottom right, there is a red banner for ".space" domains with a "Sale" tag.



Part C: DNS Enumeration

Target Domain: `nic.in`

Tool Used: Windows Command Prompt (`nslookup`)

Commands Used

DNS Record	Command
• A Record	• <code>nslookup -type=A nic.in</code>
• MX Record	• <code>nslookup -type=MX nic.in</code>
• NS Record	• <code>nslookup -type=NS nic.in</code>
• TXT Record	• <code>nslookup -type=TXT nic.in</code>

Results:-

DNS Record	Result
------------	--------

- **A Record**
- **MX Record**
- **NS Record**
- **TXT Record**
- No A record found in the query.
- **mx.mgovcloud.in** (Preference: 10), **mx2.mgovcloud.in** (Preference: 20), **mx3.mgovcloud.in** (Preference: 50)
- **ns4.nic.in**, **ns6.nic.in**, **ns8.nic.in**, **nicnet.nic.in**
- SPF record, Google Site Verification, Microsoft Verification (MS=...), and Zoho Verification records were found.

```

Command Prompt
*** No internal type for both IPv4 and IPv6 Addresses (A+AAAA) records available for nic.in
C:\Users\rmrav>nslookup -type=MX nic.in
Server: Unknown
Address: 10.143.107.3

Non-authoritative answer:
nic.in MX preference = 10, mail exchanger = mx.mgovcloud.in
nic.in MX preference = 20, mail exchanger = mx2.mgovcloud.in
nic.in MX preference = 50, mail exchanger = mx3.mgovcloud.in

C:\Users\rmrav>nslookup -type=NS nic.in
Server: Unknown
Address: 10.143.107.3

Non-authoritative answer:
nic.in nameserver = ns6.nic.in
nic.in nameserver = ns4.nic.in
nic.in nameserver = ns8.nic.in
nic.in nameserver = nicnet.nic.in

ns4.nic.in Internet address = 164.100.11.2
ns6.nic.in Internet address = 164.100.4.2
nicnet.nic.in Internet address = 164.100.105.105
ns8.nic.in AAAA IPv6 address = 2001:4408:10::11

C:\Users\rmrav>nslookup -type=TXT nic.in
Server: Unknown
Address: 10.143.107.3

Non-authoritative answer:
nic.in text =
"v=spf1 mx ip4:164.100.14.0/24 ip4:164.100.2.0/24 ip4:164.100.10.0/24 ip4:164.100.13.0/24 ip4:164.100.17.0/24 ip4:103.70.97.0/24 ip4:164.100.172.0/24 ip4:164.100.13.0/24 include:zoho.in include:mgovcloud.in include:spf.mailcloud.in ~all"
nic.in text =
"google-site-verification=zAG0LQ7cKdTU_TmdKuS3xT6vHVQHrsY8fj5EG8ho8uI"
nic.in text =
"MS=ms63829334"
nic.in text =
"zoho-verification=zB70328933.zmverify.mgovcloud.in"

```

Part D: Website Technology Identification

Target Website: <https://www.nic.in>

Tool Used: Wappalyzer (Chrome Extension)

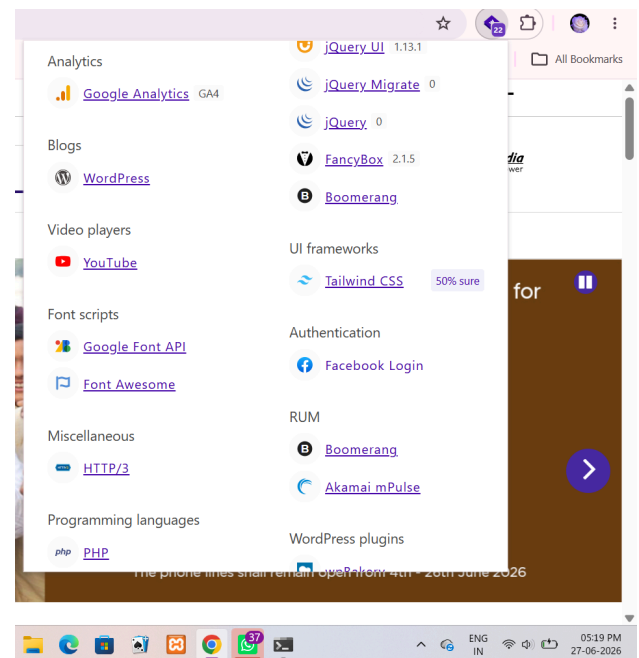
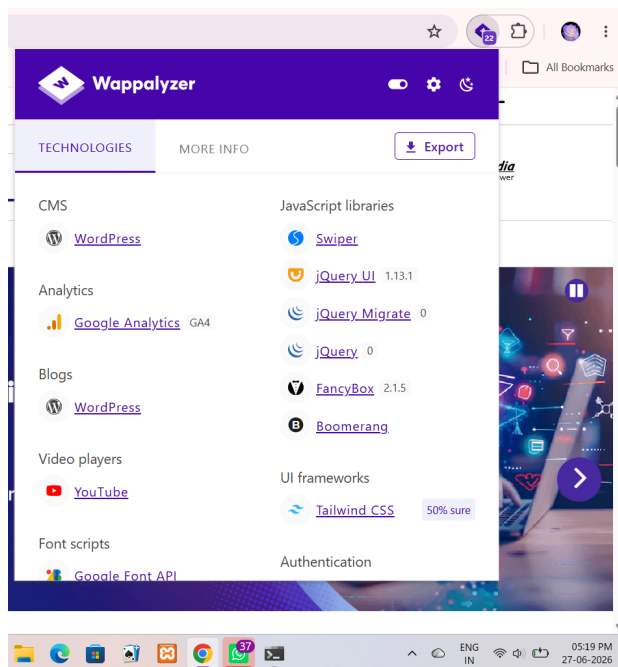
Technologies Identified

Technology Category	Technology Identified
Web Server	HTTP/3
CMS	WordPress
Programming Language	PHP

JavaScript Framework/Libraries	jQuery, jQuery UI, jQuery Migrate, Swiper, FancyBox
UI Framework	Tailwind CSS
CDN / Performance	Akamai mPulse
Analytics	Google Analytics (GA4)
Video Player	YouTube
Font Scripts	Google Font API, Font Awesome
Authentication	Facebook Login

Summary

The website technology identification was performed using the Wappalyzer browser extension. The National Informatics Centre (NIC) website uses **WordPress** as its Content Management System (CMS) and **PHP** as the programming language. It incorporates JavaScript libraries such as **jQuery**, **Swiper**, and **FancyBox** to provide interactive features. The website also uses **Tailwind CSS** for its user interface, **Google Analytics (GA4)** for visitor analytics, and **HTTP/3** for modern web communication. Additional technologies such as **Akamai mPulse**, **Google Font API**, **Font Awesome**, **YouTube**, and **Facebook Login** are integrated to improve performance and user experience.



Part E: HTTP Security Headers

Target Website: <https://www.nic.in>

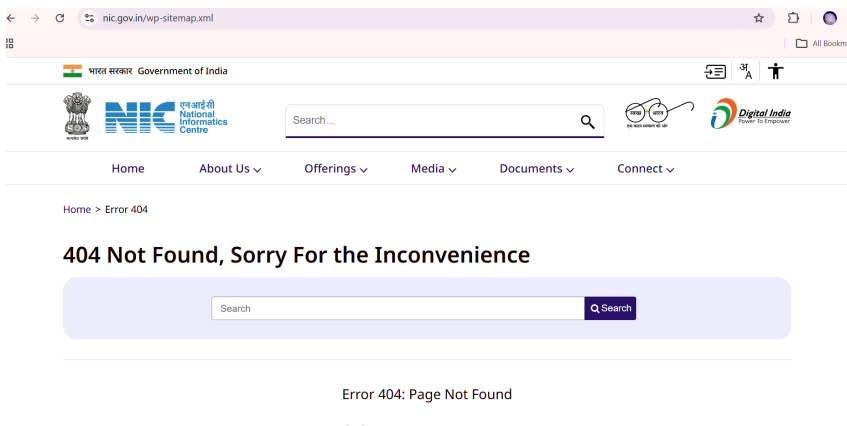
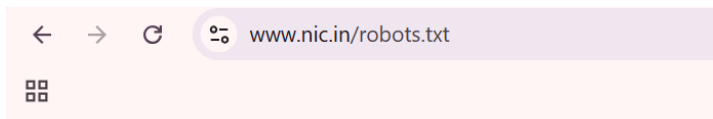
Tool Used: WebSniffer

Header	Present (Yes/No)	Purpose
Content-Security-Policy (CSP)	Yes	Restricts which resources can be loaded to help prevent Cross-Site Scripting (XSS) and code injection attacks.
X-Frame-Options	Yes	Protects the website from clickjacking attacks by preventing it from being displayed inside frames.
X-Content-Type-Options	Yes	Prevents MIME-type sniffing and ensures browsers use the correct content type.
Strict-Transport-Security (HSTS)	Yes	Forces browsers to access the website only through HTTPS.
Referrer-Policy	Yes	Controls the amount of referrer information sent with requests to other websites.

HTTP Response Header	
Name	Value
HTTP/1.1 200 OK	
Content-Type:	text/html; charset=UTF-8
Expect-CT:	enforce,max-age=2592000
Referrer-Policy:	strict-origin-when-cross-origin
X-Content-Type-Options:	nosniff
X-Frame-Options:	SAMEORIGIN
Content-Security-Policy:	img-src 'self' cbpsubscriber.mygov.in img.youtube.com *.s3waas.gov.in secure.gravatar.com data: www.nic.in informatics.nic.in personnel.nic.in xn--m1bet4hq2b.xn--h2br9c xn--m1bet4hq2b.xn--h2br9c translation-plugin.bhashini.co.in translation-plugin-stage.bhashini.co.in;connect-src 'self' *.s3waas.gov.in www.google-analytics.com translation-plugin.bhashini.co.in translation-plugin-stage.bhashini.co.in dhruva-api.bhashini.gov.in;object-src 'none';media-src 'self' *.s3waas.gov.in data;child-src *;frame-src *;form-action *.s3waas.gov.in 'self';frame-ancestors 'self' *.s3waas.gov.in ;upgrade-insecure-requests;worker-src 'self' *.s3waas.gov.in data: blob;
Referrer-Policy:	strict-origin-when-cross-origin
Permissions-Policy:	accelerometer=(),ambient-light-sensor=(),autoplay=(),battery=(),camera=(),display-capture=(),document-domain=(),encrypted-media=(),execution-while-not-rendered=(),execution-while-not-rendered=(),execution-while-out-of-viewport=(),fullscreen=(),gamepad=(),geolocation=(),magnetometer=(),gyroscope=(),magnetometer=(),layout-animations=(),legacy-image-formats=(),microphone=(),midi=(),navigation-override=(),oversized-images=(self),payment=(),picture-in-picture=(),publickey-credentials-get=(),speaker-selection=(),sync-xhr=(self),unoptimized-images=(self),unsized-media=(self),usb=(),vibrate=(),vr=(),screen-wake-lock=(),screen-wake-lock=(),web-share=(),xr-spatial-tracking=()
Strict-Transport-Security:	max-age=63072000; includeSubDomains; preload
X-Akamai-Transformed:	9 40732 0 pmb=mRUM,1
Expires:	Sat, 27 Jun 2020 12:00:00 GMT

Part F: Robots.txt & Sitemap Analysis

Question	Answer
Does the website have a robots.txt file?	Yes. The website provides a publicly accessible robots.txt file.
Does it have a sitemap?	No. A sitemap was not publicly accessible at the standard location (/sitemap.xml).
What information can be learned from these files?	The robots.txt file contains instructions for search engine crawlers, indicating which parts of the website may or may not be crawled. It helps control search engine indexing and may reference sitemap locations if available. Since no public sitemap was found, the complete list of website pages and indexing information could not be obtained from a sitemap.



Part G: Reconnaissance Report

Target Information

Field	Details
Website Name	National Informatics Centre (NIC)
Website URL	https://www.nic.in
Domain	nic.in
Target Type	Government Organization

WHOIS Results

A WHOIS lookup was performed to gather publicly available domain registration information. The domain **nic.in** is registered with the **National Informatics Centre**. The lookup provided the registration date, expiry date, domain status, and the authoritative name servers associated with the domain.

DNS Information

DNS enumeration was performed using the **nslookup** command. The results identified the domain's MX, NS, and TXT records. These records provide information about the website's mail servers, name servers, and domain verification records. No A record was found in the query performed.

Technologies Identified

Website technology identification showed that the NIC website uses **WordPress** as its Content Management System (CMS) and **PHP** as the server-side programming language. It also uses **HTTP/3**, **Tailwind CSS**, **jQuery**, **Google Analytics (GA4)**, and several other web technologies to improve performance and functionality.

Security Headers

The HTTP response headers indicate that the website implements important security mechanisms. Headers such as **Content-Security-Policy (CSP)**, **X-Frame-Options**, **X-Content-Type-Options**, **Strict-Transport-Security (HSTS)**, and **Referrer-Policy** are present, helping protect the website against common web attacks.

Robots.txt Findings

The website provides a publicly accessible **robots.txt** file, which contains instructions for search engine crawlers. However, no publicly accessible **sitemap.xml** file was found at the standard location.

Overall Observations

The reconnaissance process showed that the NIC website follows several good security practices. It uses modern web technologies, secure communication through HTTPS, properly configured DNS records, and important HTTP security headers. All information was collected using passive reconnaissance techniques without interacting with or affecting the target website.

Conclusion

During this reconnaissance activity, I learned how to gather publicly available information about a website using ethical and passive information-gathering techniques. By performing WHOIS lookup, DNS enumeration, website technology identification, HTTP security header analysis, and robots.txt inspection, I gained a better understanding of the website's infrastructure and security posture. The National Informatics Centre (NIC) website uses modern technologies and implements several security best practices, including important HTTP security headers and secure DNS configuration. This practical demonstrated that valuable information can be

collected without performing any intrusive actions. Overall, the exercise improved my understanding of the reconnaissance phase, which is an essential first step in cybersecurity assessments and penetration testing.